

could help paralysed patients move their arms and legs.

The fuel cell consists of a platinum catalyst that strips electrons from glucose to create a small electric current, mimicking the activity of cellular enzymes that break down glucose to generate ATP (the cell's energy currency). Platinum has a proven record of long-term biocompatibility within the body.

Led by Rahul Sarpeshkar, an associate professor of electrical engineering and computer science at MIT, the researchers fabricated the fuel cell on a silicon chip so that it could interface with other circuits in a brain implant. The fuel cell is fabricated from silicon using the same technology as used to make semiconductor electronic chips. So far, it can generate up to hundreds of microwatts—enough to power neural implants used to help paralysed patients move their limbs.

New technology converts ink into sound

We have seen flexible electronic circuits from Xerox and also flexible organic flash memory from Tokyo University in the past. But it seems that a German researcher could be the first to commercialise electronics printed on a sheet of paper. Professor Hübner, who runs an R&D group at the Chemnitz University (pmTUC) in Germany, has developed a technology that converts ink into sound. The newly developed ink makes it possible to produce flat speakers or printed solar cells.

The printed paper loudspeaker is connected to an audio amplifier like a conventional loudspeaker. "Frequency response and hence sound quality is very good and the paper is surprisingly loud. Just the bass of the paper-based loudspeaker is a bit weak," explains Dr Georg Schmidt, senior researcher at pmTUC. The thin loudspeakers, which are printed in the laboratories of pmTUC, contain several layers of a conductive organic polymer and a piezoelectric layer.

According to project assistant Maxi Bellmann, the loudspeakers can be produced in a very cheap way as mass printing methods are used. It is possible to print coloured messages on unused space available at the bottom side of the paper loudspeaker.

Faster photodetector created by tricking graphene

Researchers at the Center for Nanophysics and Advanced Materials, University of Maryland, have developed a new type of hot-electron bolometer that acts as a sensitive detector of infrared light. It can be used in applications ranging from detection of chemical and biochemical weapons from a distance to chemical analysis in the laboratory and studying the structure of the universe through improved telescopes.

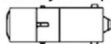
The researchers developed the bolometer using bilayer graphene—two-atom-thick sheets of carbon. The bolometer is likely to be sensitive to a very broad range of light energies, ranging from terahertz frequencies or submillimetre waves through the infrared to visible light.

Graphene, a single-atom-thick plane of graphite, has some unique properties. Due to a bandgap of exactly zero energy, it can absorb photons of any energy. This means graphene can

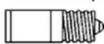
BINAY Multichip LED Lamps

These multichip LED lamps are ideal for use in applications such as annunciator window fascia lighting and illuminated pushbutton switches

BA9S bayonet cap



E10 screw cap



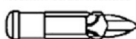
Midget Groove



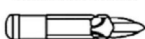
Midget Flange



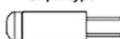
5.5mm slide contact



6.8mm slide contact



Bi-pin type



Wedge Base



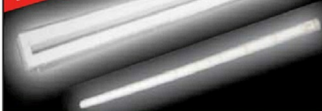
Also available in E5, E12, E14, and B14 base types



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